Clasa a 7-a / Formule si Teoreme Esentiale

I. Multimea Numerelor Reale

- $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ (incluziunile multimilor de numere)
- Irationale: $\sqrt{2}$, $\sqrt{3}$, π etc. (zecimale infinite, fara repetitie)
- $\sqrt{n}$ rational <=> n este patrat perfect

Estimare: $n^2 < a < (n+1)^2 \Rightarrow n < \sqrt{a} < n+1$

Scoatere factori: $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$

Introducere: $a\sqrt{b} = \sqrt{(a^2 \times b)}$

- Comparare: introducem sub radical si comparam ($3\sqrt{2}=\sqrt{18}$ vs $2\sqrt{5}=\sqrt{20}$)

II. Operatii cu Radicali

$\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$; $\sqrt{a} \div \sqrt{b} = \sqrt{(a/b)}$

$(\sqrt{a})^2 = a$; $\sqrt{(a^2)} = |a|$

- Radicali asemenea = acelasi numar sub radical => se aduna/scad

$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$ (ca $3a + 5a = 8a$)

- Neasemanatori NU se pot simplifica: $\sqrt{2} + \sqrt{3}$ ramane asa

III. Rationalizarea Numitorului

Numitor $\sqrt{b}$: inmultim cu $\sqrt{b}/\sqrt{b}$

$1/\sqrt{2} = \sqrt{2}/2$; $3/\sqrt{5} = 3\sqrt{5}/5$

Numitor $a+\sqrt{b}$: inmultim cu conjugatul $(a-\sqrt{b})$

$(a+b)(a-b) = a^2-b^2$ => $(\sqrt{5}+1)(\sqrt{5}-1) = 4$

- $1/(1+\sqrt{2}) = (1-\sqrt{2})/((1+\sqrt{2})(1-\sqrt{2})) = (1-\sqrt{2})/(1-2) = \sqrt{2}-1$

IV. Puteri cu Exponent Intreg

a^0 = 1 ($a \neq 0$); a^(-n) = 1/a^n

a^m $\times$ a^n = a^(m+n); a^m $\div$ a^n = a^(m-n)

(a^m)^n = a^(mn); (ab)^n = a^n b^n; (a/b)^n = a^n/b^n

V. Calcul Algebric

$(a+b)^2 = a^2+2ab+b^2$; $(a-b)^2 = a^2-2ab+b^2$

$(a+b)(a-b) = a^2-b^2$

$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$

$a^3-b^3 = (a-b)(a^2+ab+b^2)$

- Factorizare: scoatere factor comun, formule remarcabile, grupare

VI. Ecuatii si Inecuatii

- Ecuatia de gradul I: $ax+b=0 \Rightarrow x=-b/a$
- La inecuatii cu numar NEGATIV: semnul se INVERSEAZA
- Sisteme: substitutie sau reducere

VII. Geometrie - Triunghiul

Suma unghiurilor = 180°; Unghi exterior = suma celor 2 nealaturate

Pitagora: $c^2 = a^2 + b^2$; Triplete: (3,4,5), (5,12,13), (8,15,17)

- Arie = baza x h / 2; P = a+b+c
- Isoscel: unghiurile bazei egale; Echilateral: laturi egale, unghiuri 60
- Congruenta: LLL, LUL, ULU, LLU
- Relatii metrici: in isoscel mediatoarea = bisectoarea = inaltime din varf

VIII. Patrulaterale speciale

- Patrat: $A=l^2$, $d=l\sqrt{2}$; Dreptunghi: $A=L \times l$, $d=\sqrt{L^2+l^2}$
- Romb: $A=d1 \times d2/2$; Paralelogram: $A=a \times h$
- Trapez: $A=(B+b) \times h/2$; Trapez isoscel: diagonalele egale

IX. Cercul

$C=2\pi r \approx 6{,}28r$; $A=\pi r^2 \approx 3{,}14r^2$

- Tangenta la cerc perpendiculara pe raza in punctul de tangenta
- Lungime arc = $(\angle AOB/360) \times 2\pi r$; Arie sector = $(\angle AOB/360) \times \pi r^2$

X. Corpuri Geometrice - Arie si Volum

Cub: $A=6a^2$; $V=a^3$; $diag=\sqrt{3} \times a$

Cuboid: $A=2(ab+bc+ca)$; $V=a\times b\times c$

Cilindru: A_lat=$2\pi rh$; A_tot=$2\pi r(r+h)$; $V=\pi r^2 h$

Con: A_lat=πrl (l=gen.); $V=\pi r^2 h/3$

Sfera: $A=4\pi r^2$; $V=4\pi r^3/3$

Piramida: V=A_baza x h / 3

XI. Trigonometrie in triunghiul dreptunghic

sin $\angle A$ = cateta opusa / ipotenuza

cos $\angle A$ = cateta alaturata / ipotenuza

tg $\angle A$ = cateta opusa / cateta alaturata

$\sin^2\angle A + \cos^2\angle A = 1$ (identitate fundamentala)

- sin30=1/2, cos30=$\sqrt{3}/2$, tg30=$\sqrt{3}/3$
- sin45=$\sqrt{2}/2$, cos45=$\sqrt{2}/2$, tg45=1
- sin60=$\sqrt{3}/2$, cos60=1/2, tg60=$\sqrt{3}$

XII. Statistici si Probabilitati

Media aritmetica = suma(xi x ni) / N

Dispersia = suma((xi - media)^2 x ni) / N

P(A) = card(A) / card(Omega)

- P(A+B) = P(A) + P(B) - P(A x B)

Radicali:

- $\sqrt{12} + \sqrt{27}$ = ___ ; $\sqrt{50} - \sqrt{18}$ = ___ ; $3\sqrt{2} \times 2\sqrt{8}$ = ___
- $(\sqrt{5}+2)(\sqrt{5}-2)$ = ___ ; $(\sqrt{3}+1)^2$ = ___
- $1/\sqrt{3}$ = ___ ; $2/(1+\sqrt{5})$ = ___ (rationalizeaza)

Puteri:

- 2^3 × 2^(-3) = ___ ; (3^2)^3 = ___ ; 5^(-2) = ___
- 2^3 + 4× 2^(-1) - $\sqrt{16}$ = ___

Geometrie:

- Triunghi isoscel: $\angle A=40°$ => $\angle B=\angle C$ = ___
- Pitagora: a=5, b=12 => c=___ ; cateta isoscel=5, ipoten.=$5\sqrt{2}$ => cateta=___
- Cub cu a=3: A=___ ; V=___
- Cilindru r=2, h=5: V≈___

Trigonometrie:

- In triunghi dreptunghic, ipot=10, $\angle A=30°$: cateta opusa=___, cateta alaturata=___
- $\sin^2 30 + \cos^2 30$ = ___

Calcul algebric:

- $(x+3)^2$ = ___ ; $(x+5)(x-5)$ = ___ ; $(2x-1)^2$ = ___
- Factorizeaza: x^2-9 = ___ ; $4x^2-4x+1$ = ___